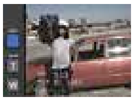
Test it!

iPhone users can now 'Test drive' Samsung's new phones for 30 days for just $1.

http://dgit.in/testsmsngphne

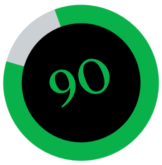


Go retro!

Go retro with this app, which gives a 1985 camcorder feel while recording on an iOS device. Check it out: **http://dgit.in/ioscmcrdrapp**

Intel Core i7-6700K

Skylake makes a splash



Performance................ 90

Specifications

Clock speed: 4.0 GHz, 4.2 GHz on Turbo; **Cache:** 8 MB; **Cores:** 4; **Threads:** 8; **Manufacturing process:** 14 nm FinFET; **TDP:** 91W; **IGP:** HD Graphics 530; **IGP Clock Speed**: 350 MHz, 1150 MHz on Turbo; **Memory:** Dual Channel DDR4 or DDR3L; **PCIe Lanes:** 16 ver 3.0; **Socket:** LGA 1151

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Let's get the whole Tick-Tock bit out of the way. Skylake is the successor to Broadwell which was released just a few months back. It's what Intel describes as a "tock" or an improvement in the microarchitecture design. Ideally, followed by a "Tick" that shrinks the design, Intel will opt for another tock since the 10nm manufacturing process is taking a little more time to mature. Along with the Core i7 6700K, Intel has decided to release just one other SKU, i.e. the Core i5 6600K with more to come later this year.

The Core i7 6700K is clocked at 4.0 GHz with a turbo clock that pushes it be a meagre 0.2 GHz. In comparison, the Core i5 6600K ends up with a much higher performance delta. Compared to Broadwell and Devil's Canyon, this new architecture doesn't really push the envelope. Generally, we're used to seeing increments of around 9-11 per cent between generations but with the gap between Intel's releases narrowing with each year, this kind of behaviour was expected.

Skylake also comes with a brand new socket – The LGA 1151. And this makes it incompatible with previous generation of motherboards i.e. the 87-series and 97-series. This isn't just because of the extra pin but for the fact that Intel decided to take out the voltage regulator off the die and back into the hands of motherboard manufacturers. FIVR (Fully Integrated Voltage Regulator) was introduced to give the user a much finer control over the CPU voltage. If you've ever looked at a motherboard then the huge metal heatsinks around the processor are basically there to cool down the voltage regulator circuitry. Imagine the effect of putting this on the CPU die. Yes, it led to more heating when users had to overclock the CPU.

Reducing the manufacturing process down to the 14nm FinFET process also reduces the voltage level required to drive the circuitry, which means that Skylake gets DDR4 support. Also, since DDR3L (L-Low Power) runs at 1.35 volts which is quite close to the processor's 1.2 volts, it too will be compatible. DDR4's high cost might put off a lot of enthusiasts and lead them to opt for DDR3L supported motherboards. This incurs a performance drop if the DDR3L is to be slower than 1866 MHz and the ironic thing about these SKUs are that they cost pretty much the same as DDR4 DIMMs clocked at 2133 MHz so you might as well opt for DDR4 running at close to native speeds.

The integrated GPU with Skylake has undergone a change in nomenclature with Intel having ditched the 4-digit system for a 3-digit one. The IGP on the Core i7 6700K is called HD Graphics 530 and sports 24 Execution Units clocked at 350 MHz on idle while taking it all the way up to 1150 MHz on turbo. Performance wise, it can deliver about 40 FPS in current games like Bioshock Infinite which puts it at par with the AMD R7 240.

Performance wise, Skylake's Core i7 6700K ends up being slightly greater than Devil's Canyon (Core i7 4790K). This increment of 4-5 per cent can barely justify upgrading your PC and this has been the tale for quite a while. Which makes one wonder who'd best benefit from upgrading to Skylake, well if you happen to have a 4770K or a 3770K or even a 2600K then you might want to wait another year. Sandy Bridge is sort of where you can draw the line, the performance increment over Sandy Bridge and the bevy of new features that the Z170 chip brings to the table makes it a considerable upgrade. Prime among these would be M.2 and SATA Express with native NVMe support. We've already seen motherboards on which you can easily obtain data transfer speeds of over 3300 MB/s using three M.2 SSDs in RAID. What makes an upgrade even more enticing is that you can still get a decent resale value for your old Sandy Bridge PC.

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